Exact PTO Equivalence Decimal Multiplication

Metric-Extracted Controller

`analyses/metric_extraction/DFST/mult10/warmup_0_steps_4000000/`

1 Statement and Scope

Controllers. The export `minimized_partial.json` defines controller C, with 182 states, initial state 0, and halt state 181. The function `mult_grid(10)` defines expert $\mathcal{E}$, with 108 states, initial state 0, and halt state 107. The adjacent expert export matches this function entry for entry. All state numbers below follow the original exports.

Input Domain. Let $D = \{0, 1, \ldots, 9\}$. Row 0 initially contains the contiguous word $a\,x\,b$, starting at column 0, where $a \in D^m$, $b \in D^n$, and $m, n \geq 1$. All remaining grid cells contain blanks. The head starts at $(0, 0)$. Write $A = \mathrm{val}_{10}(a)$ and $B = \mathrm{val}_{10}(b)$. The domain permits arbitrary finite operand lengths, unequal lengths, leading zeros, and zero-valued operands.

Execution Convention. The grid uses $\mathbb{Z}^2$, with coordinates increasing rightward and downward. The symbol λ denotes a blank. A transition (q', w, M) writes w, performs motion M, and enters state q'. Motions L, R, D, U move one cell; motion S writes and stops without moving. An undefined transition would prevent continuation.

Policy-Trajectory Observations. Define the PTO on input $a\,x\,b$ by

$$\mathrm{PTO}(a\,x\,b) = \big((r_t, w_t, M_t)\big)_{t=0}^{T-1},$$

where r_t, w_t, M_t denote the read symbol, written symbol, and motion at step t. Count the final stopping action exactly once, without halt-loop padding.

Theorem 1 (Exact PTO Equivalence)**.** *For every input in the stated domain,*

$$\mathrm{PTO}_C(a\,x\,b) = \mathrm{PTO}_{\mathcal{E}}(a\,x\,b).$$

Both executions terminate on the same step. The complete grids and head positions agree at every step. No undefined controller entry occurs. The final grid contains only the $(m + n)$-digit decimal representation of AB, with leading-zero padding, on row $n - 1$ in columns $-n, \ldots, m - 1$. The final head stops at $(-n - 1, n - 1)$.

The theorem concerns the exported discrete controller. The proof combines geometric and arithmetic invariants with a finite simulation certificate; no bound on m or n enters the argument.

2 Expert Behavior

2.1 Copying and Partial Products

Write $b = b_{n-1} \cdots b_0$, so $B = \sum_{i=0}^{n-1} b_i 10^i$. Assign global weight 10^{m-1-z} to column z. Column $m-1$ carries the units digit of the final product.

Lemma 1 (Copy Staircase)**.** *Before product formation, the expert constructs exactly n occupied rows. For $0 \le i < n$, row i contains*

$$a\, x\, b_{n-1} \cdots b_i \quad \text{starting in column } -i. \tag{1}$$

All remaining cells contain blanks.

Proof. Row 0 contains the input. On row i, the expert scans to the right blank, steps left over b_i, and inspects the preceding cell. A digit at this position implies at least two remaining factor digits. The expert copies every symbol except b_i one row down and one column left, from right to left. Each destination contains a blank before writing; the source row remains unchanged.

The first descent reaches a blank. Subsequent descents reach the copied right neighbor: a digit, except when copying the last digit of a, where the lower cell contains the copied x. The descent from the source x reaches a digit. After copying the first digit of a, the expert reads the left blank and descends onto the first digit of the new row. The new row satisfies (1), with one fewer factor digit.

When one factor digit remains, the second left step reaches x and starts product formation. Thus $n-1$ copies produce exactly the stated staircase. □

Lemma 2 (Aligned Partial Products)**.** *After product formation, row i contains the $(m+1)$-digit representation of Ab_i on columns $-i-1, \ldots, m-i-1$. The leading digit belongs to $\{0, \ldots, 8\}$. In global column weights, this row represents $Ab_i 10^i$. All remaining cells contain blanks.*

Proof. The expert processes rows from $n-1$ upwards. On row i, the expert erases the selected factor $f = b_i$, the remaining factor prefix, and x, then walks left across a. For $f = 0$, each digit becomes 0, followed by one padding 0 on the left blank. For $1 \le f \le 9$, the control state stores carry c, initially 0, and applies

$$fd + c = w + 10c', \qquad w = (fd + c) \bmod 10, \quad c' = \lfloor (fd + c)/10 \rfloor. \tag{2}$$

The bound $0 \le c < f$ persists because $fd + c \le 9f + f - 1 = 10f - 1$. The usual suffix invariant for (2) gives the product digits; the left blank receives the final carry. Thus the leading digit never exceeds $f - 1 \le 8$.

The right scan ends at column $m - i$ and moves up. For $i > 0$, the upper cell contains the last digit of a, followed by x and the remaining factor prefix. The expert scans this prefix, selects the last digit, and repeats. Every higher row contains at least two factor digits, so at least one further digit follows the selected factor during leftward erasure. For $i = 0$, the upper cell contains a blank, starting summation on row 0. □

Every scan crosses a finite word. Copying shortens the factor prefix; product formation visits each occupied row once. Both passes terminate.

2.2 Accumulation and Carry Confinement

Lemma 3 (Fixed-Width Accumulator)**.** *After merging through row k, $0 \le k < n$, all rows above k contain blanks. Row k contains a contiguous $(m + k + 1)$-digit accumulator on columns $-k - 1, \ldots, m - 1$, with value*

$$S_k = A \sum_{i=0}^{k} b_i 10^i.$$

Rows below k retain the partial products from Lemma 2. At inspection of the next row, expert state 88 has head $(-k - 2, k + 1)$.

Proof. For $k = 0$, the expert scans left across the product on row 0, reaches column -2, and descends to inspect row 1. If row $k + 1$ exists, inspection reads the fresh leading digit in $\{0, \ldots, 8\}$. The expert scans the lower product to the right blank at $m - k - 1$ and moves up. The upper cell contains an accumulator digit, since

$$-k - 1 \le m - k - 1 \le m - 1.$$

The expert scans to the upper right blank and steps left to column $m - 1$. The lower product ends at $m - k - 2$, so the first lower cell contains a blank.

For upper digit u, lower digit v (a blank contributes zero), and incoming carry $c \in \{0, 1\}$, the update applies

$$u + v + c = w + 10c', \qquad w \in D, \quad c' \in \{0, 1\}. \tag{3}$$

The expert erases u, writes w below, returns to the erased upper cell, and moves left. The processed lower suffix contains the correct sum suffix; the unprocessed cells retain the old digits; c' records the contribution to the next column. Equation (3) preserves this invariant.

The lower word extends one column farther left than the upper word. After the upper word ends, carry zero preserves this lower leading digit. Carry one increments the leading digit and clears, since the digit never exceeds 8. The expert reaches column $-k - 3$ on row $k + 1$ and descends. Row k contains only blanks; row $k + 1$ represents $S_k + Ab_{k+1}10^{k+1} = S_{k+1}$ on the claimed support. □

Lemma 4 (Boundary Restrictions)**.** *A carry-one update beneath an upper digit reads an original lower digit. A blank lower update guarantees another upper digit on the next left step. A carry beyond the upper word reads a lower digit in $\{0, \ldots, 8\}$ and clears.*

Proof. The blank gap lies strictly to the right of the original lower word. The first update has carry zero, and $u + 0 \le 9$, so no carry originates inside this gap. Carry one can first arise only inside the lower digit block; further leftward updates stay inside that block until the leading digit.

A blank lower cell has column $z \ge m - k - 1$. The next upper column satisfies $z - 1 \ge m - k - 2 \ge -k - 1$, using $m \ge 1$, and therefore still contains a digit. Finally, the lower leading digit lies one column left of the upper word. Lemma 2 bounds this untouched digit by 8; adding 1 clears any final carry without extending the fixed-width support. □

Each merge consumes finitely many upper digits and at most one extra carry column. After $n - 1$ merges, state 88 reads the blank below the final row, moves up to state 106, and emits S into halt state 107. Lemma 3 gives the final product, padding, and head position.

3 Observation Protocol

Section 4 gives 207 phases. Phase i pairs expert state g_i with controller state $q \in Q_i$, on an identical grid and head. An edge $i \xrightarrow{x} j$ permits read symbol x and advances to phase j. The protocol begins at phase 0 and ends at phase 206. All phase sets contain one controller state except $Q_{201} = \{26, 106\}$.

Lemma 5 (Protocol Coverage)**.** *Every expert execution in the stated domain follows the protocol in Section 4 from phase* 0 *to phase* 206*.*

Proof. The following geometric and arithmetic conditions justify every read restriction.

Copying. The initial scan and every restart encounter a nonempty left operand, then x, then a nonempty right operand before the right blank. The second left step reads a factor digit or x. Lemma 1 distinguishes the first descent (blank), ordinary descents (digits), and the descent directly above the copied operator (x). Every destination contains a blank. After the first copied symbol, the next source symbol contains a digit or x; no left blank can occur yet. After copying x, the next source symbol contains the last digit of a. The ordinary copy loop eventually reaches the left blank and restarts.

Products. The bottom row contains one factor digit; erasing that digit immediately exposes x. Every higher row contains at least two factor digits; erasing the selected factor therefore exposes another digit before x. After erasing x, the first product update reads a digit because $m \geq 1$. Later updates read digits followed by the left blank. Equation (2) determines the next carry phase and preserves the carry bound. The completed product scan reads digits followed by a blank. Lemma 2 places the head on an upper digit followed by x and a nonempty factor prefix, or on a blank above row 0.

Alignment and Summation. Summation starts on the right blank of row 0. The initial left scan crosses a nonempty product. Inspection below reads a fresh leading digit from $\{0, \ldots, 8\}$ or a blank below the last row. After a leading digit, at least one lower digit remains because $m \geq 1$. Lemma 3 places the first upper alignment step on a digit; further alignment steps read digits until the right blank. The first upper units digit has carry zero and a blank lower cell.

After every lower update, the upper return cell contains the erased blank. A blank lower update leads to a nonblank upper read by Lemma 4. An update inside the lower digit block leads to an upper digit or the left blank. Carry-one digit updates remain inside the lower digit block. At the end of the upper word, carry zero preserves the untouched lower leading digit, while carry one increments that digit and clears. Both paths reach the blank immediately left of the merged word. The final stop reads a blank.

These conditions give exactly the permitted branches used by the expert. Induction over individual moves, copied rows, and merged rows establishes coverage. The finite-pass arguments in Lemmas 1–3 give termination at phase 206. □

The protocol may permit additional observation sequences. Checking every permitted branch therefore suffices for all valid grid executions.

4 Simulation Certificate

Finite Local Certificate. For every protocol edge $i \xrightarrow{x} j$ and every $q \in Q_i$, direct comparison with the two exports gives

$$\delta_{\mathcal{E}}(g_i, x) = (g_j, w, M), \qquad \delta_C(q, x) = (q', w, M), \qquad q' \in Q_j. \tag{4}$$

Every required entry exists. A transition reaches halt exactly when $j = 206$, with motion S. The companion checker verifies all 1,509 obligations across 208 phase–state pairs. These obligations cover all 1,259 defined nonterminal controller entries. The remaining 12 defined entries contain halt loops, which never contribute to a PTO.

4.1 Phases and Controller-State Relation

The table gives the phase index i, expert state g_i, controller set Q_i, and permitted reads with successor phase indices. For a digit vector $[j_0, \ldots, j_9]$, read d advances to j_d. The notation $D : j$ sends every decimal digit to phase j; $0{:}8 : j$ permits only digits $0, \ldots, 8$. Unlisted reads never occur in the corresponding phase of a valid input.

The expert transition supplies the write and motion for each edge; Equation (4) checks both exports against that action. The copying moves follow Lemma 1; product updates follow (2); addition updates follow (3). Phase labels describe positions before the next action.

i	Phase	g_i	Q_i	Read : next phase
0	First left-operand digit	0	{0}	$D : 1$
1	Scan left operand	0	{0}	$D : 1 \quad x : 2$
2	First factor digit	0	{1}	$D : 3$
3	Scan right operand	0	{1}	$D : 3 \quad \lambda : 4$
4	Skip last factor digit	1	{2}	$D : 5$
5	Copy or start products	2	{3}	$D : [6, 8, 10, 12, 14, 16, 18, 20, 22, 24]$ $x : 62$
6	First descent, digit 0	3	{172}	$\lambda : 7$
7	First copied write, digit 0	14	{173}	$\lambda : 26$
8	First descent, digit 1	4	{123}	$\lambda : 9$
9	First copied write, digit 1	15	{124}	$\lambda : 26$
10	First descent, digit 2	5	{64}	$\lambda : 11$
11	First copied write, digit 2	16	{65}	$\lambda : 26$
12	First descent, digit 3	6	{4}	$\lambda : 13$
13	First copied write, digit 3	17	{5}	$\lambda : 26$
14	First descent, digit 4	7	{87}	$\lambda : 15$
15	First copied write, digit 4	18	{88}	$\lambda : 26$
16	First descent, digit 5	8	{39}	$\lambda : 17$
17	First copied write, digit 5	19	{40}	$\lambda : 26$
18	First descent, digit 6	9	{163}	$\lambda : 19$
19	First copied write, digit 6	20	{164}	$\lambda : 26$
20	First descent, digit 7	10	{51}	$\lambda : 21$
21	First copied write, digit 7	21	{52}	$\lambda : 26$
22	First descent, digit 8	11	{128}	$\lambda : 23$
23	First copied write, digit 8	22	{129}	$\lambda : 26$
24	First descent, digit 9	12	{79}	$\lambda : 25$
25	First copied write, digit 9	23	{80}	$\lambda : 26$
26	Source after first copy	25	{6}	$D : [27, 29, 31, 33, 35, 37, 39, 41, 43, 45]$ $x : 48$
27	Ordinary descent, digit 0	3	{178}	$D : 28$
28	Ordinary copied write, digit 0	14	{159}	$\lambda : 47$
29	Ordinary descent, digit 1	4	{134}	$D : 30$

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Simulation Certificate (Continued)

i	Phase	g_i	Q_i	Read : next phase
30	Ordinary copied write, digit 1	15	{135}	λ : 47
31	Ordinary descent, digit 2	5	{68}	D : 32
32	Ordinary copied write, digit 2	16	{67}	λ : 47
33	Ordinary descent, digit 3	6	{125}	D : 34
34	Ordinary copied write, digit 3	17	{11}	λ : 47
35	Ordinary descent, digit 4	7	{166}	D : 36
36	Ordinary copied write, digit 4	18	{90}	λ : 47
37	Ordinary descent, digit 5	8	{165}	D : 38
38	Ordinary copied write, digit 5	19	{131}	λ : 47
39	Ordinary descent, digit 6	9	{55}	D : 40
40	Ordinary copied write, digit 6	20	{42}	λ : 47
41	Ordinary descent, digit 7	10	{170}	D : 42
42	Ordinary copied write, digit 7	21	{148}	λ : 47
43	Ordinary descent, digit 8	11	{150}	D : 44
44	Ordinary copied write, digit 8	22	{116}	λ : 47
45	Ordinary descent, digit 9	12	{167}	D : 46
46	Ordinary copied write, digit 9	23	{54}	λ : 47
47	Continue source copy	25	{12}	D : [27, 29, 31, 33, 35, 37, 39, 41, 43, 45] x : 48 λ : 61
48	Descend below operator	13	{7}	D : 49
49	Write copied operator	24	{8}	λ : 50
50	Copy last digit of a	25	{9}	D : [51, 52, 53, 54, 55, 56, 57, 58, 59, 60]
51	Descend above x, digit 0	3	{158}	x : 28
52	Descend above x, digit 1	4	{145}	x : 30
53	Descend above x, digit 2	5	{66}	x : 32
54	Descend above x, digit 3	6	{10}	x : 34
55	Descend above x, digit 4	7	{89}	x : 36
56	Descend above x, digit 5	8	{130}	x : 38
57	Descend above x, digit 6	9	{41}	x : 40
58	Descend above x, digit 7	10	{147}	x : 42
59	Descend above x, digit 8	11	{115}	x : 44
60	Descend above x, digit 9	12	{53}	x : 46
61	Restart on copied row	0	{13}	D : 1
62	Bottom-row factor	26	{14}	D : [63, 66, 69, 73, 78, 84, 91, 99, 108, 118]
63	Erase x, bottom factor 0	27	{111}	x : 64
64	First product digit, factor 0	37	{62}	D : 65
65	Continue product by zero	37	{62}	D : 65 λ : 129
66	Erase x, bottom factor 1	28	{78}	x : 67
67	First product digit, factor 1	38	{24}	D : 68
68	Product by 1, carry 0	38	{24}	D : 68 λ : 129
69	Erase x, bottom factor 2	29	{69}	x : 70
70	First product digit, factor 2	39	{70}	D : [71, 71, 71, 71, 71, 72, 72, 72, 72, 72]
71	Product by 2, carry 0	39	{70}	D : [71, 71, 71, 71, 71, 72, 72, 72, 72, 72] λ : 129
72	Product by 2, carry 1	40	{143}	D : [71, 71, 71, 71, 71, 72, 72, 72, 72, 72] λ : 129
73	Erase x, bottom factor 3	30	{15}	x : 74
74	First product digit, factor 3	41	{16}	D : [75, 75, 75, 75, 76, 76, 76, 77, 77, 77]
75	Product by 3, carry 0	41	{16}	D : [75, 75, 75, 75, 76, 76, 76, 77, 77, 77] λ : 129
76	Product by 3, carry 1	42	{48}	D : [75, 75, 75, 76, 76, 76, 76, 77, 77, 77] λ : 129
77	Product by 3, carry 2	43	{155}	D : [75, 75, 75, 76, 76, 76, 77, 77, 77, 77] λ : 129
78	Erase x, bottom factor 4	31	{75}	x : 79

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Simulation Certificate (Continued)

i	Phase	g_i	Q_i	Read : next phase
79	First product digit, factor 4	44	{76}	D : [80, 80, 80, 81, 81, 82, 82, 82, 83, 83]
80	Product by 4, carry 0	44	{76}	D : [80, 80, 80, 81, 81, 82, 82, 82, 83, 83] λ : 129
81	Product by 4, carry 1	45	{91}	D : [80, 80, 80, 81, 81, 82, 82, 82, 83, 83] λ : 129
82	Product by 4, carry 2	46	{77}	D : [80, 80, 81, 81, 81, 82, 82, 83, 83, 83] λ : 129
83	Product by 4, carry 3	47	{117}	D : [80, 80, 81, 81, 81, 82, 82, 83, 83, 83] λ : 129
84	Erase x, bottom factor 5	32	{43}	x : 85
85	First product digit, factor 5	48	{44}	D : [86, 86, 87, 87, 88, 88, 89, 89, 90, 90]
86	Product by 5, carry 0	48	{44}	D : [86, 86, 87, 87, 88, 88, 89, 89, 90, 90] λ : 129
87	Product by 5, carry 1	49	{153}	D : [86, 86, 87, 87, 88, 88, 89, 89, 90, 90] λ : 129
88	Product by 5, carry 2	50	{144}	D : [86, 86, 87, 87, 88, 88, 89, 89, 90, 90] λ : 129
89	Product by 5, carry 3	51	{45}	D : [86, 86, 87, 87, 88, 88, 89, 89, 90, 90] λ : 129
90	Product by 5, carry 4	52	{95}	D : [86, 86, 87, 87, 88, 88, 89, 89, 90, 90] λ : 129
91	Erase x, bottom factor 6	33	{107}	x : 92
92	First product digit, factor 6	53	{108}	D : [93, 93, 94, 94, 95, 96, 96, 97, 97, 98]
93	Product by 6, carry 0	53	{108}	D : [93, 93, 94, 94, 95, 96, 96, 97, 97, 98] λ : 129
94	Product by 6, carry 1	54	{171}	D : [93, 93, 94, 94, 95, 96, 96, 97, 97, 98] λ : 129
95	Product by 6, carry 2	55	{110}	D : [93, 93, 94, 95, 95, 96, 96, 97, 98, 98] λ : 129
96	Product by 6, carry 3	56	{174}	D : [93, 93, 94, 95, 95, 96, 96, 97, 98, 98] λ : 129
97	Product by 6, carry 4	57	{142}	D : [93, 94, 94, 95, 95, 96, 97, 97, 98, 98] λ : 129
98	Product by 6, carry 5	58	{109}	D : [93, 94, 94, 95, 95, 96, 97, 97, 98, 98] λ : 129
99	Erase x, bottom factor 7	34	{56}	x : 100
100	First product digit, factor 7	59	{57}	D : [101, 101, 102, 103, 103, 104, 105, 105, 106, 107]
101	Product by 7, carry 0	59	{57}	D : [101, 101, 102, 103, 103, 104, 105, 105, 106, 107] λ : 129
102	Product by 7, carry 1	60	{168}	D : [101, 101, 102, 103, 103, 104, 105, 106, 106, 107] λ : 129
103	Product by 7, carry 2	61	{127}	D : [101, 101, 102, 103, 104, 104, 105, 106, 106, 107] λ : 129
104	Product by 7, carry 3	62	{179}	D : [101, 102, 102, 103, 104, 104, 105, 106, 106, 107] λ : 129
105	Product by 7, carry 4	63	{59}	D : [101, 102, 102, 103, 104, 104, 105, 106, 107, 107] λ : 129
106	Product by 7, carry 5	64	{169}	D : [101, 102, 102, 103, 104, 105, 105, 106, 107, 107] λ : 129
107	Product by 7, carry 6	65	{58}	D : [101, 102, 103, 103, 104, 105, 105, 106, 107, 107] λ : 129
108	Erase x, bottom factor 8	35	{112}	x : 109
109	First product digit, factor 8	66	{113}	D : [110, 110, 111, 112, 113, 114, 114, 115, 116, 117]
110	Product by 8, carry 0	66	{113}	D : [110, 110, 111, 112, 113, 114, 114, 115, 116, 117] λ : 129
111	Product by 8, carry 1	67	{137}	D : [110, 110, 111, 112, 113, 114, 114, 115, 116, 117] λ : 129

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Simulation Certificate (Continued)

i	Phase	g_i	Q_i	Read : next phase
112	Product by 8, carry 2	68	{121}	D : [110, 111, 111, 112, 113, 114, 115, 115, 116, 117] λ : 129
113	Product by 8, carry 3	69	{114}	D : [110, 111, 111, 112, 113, 114, 115, 115, 116, 117] λ : 129
114	Product by 8, carry 4	70	{132}	D : [110, 111, 112, 112, 113, 114, 115, 116, 116, 117] λ : 129
115	Product by 8, carry 5	71	{149}	D : [110, 111, 112, 112, 113, 114, 115, 116, 116, 117] λ : 129
116	Product by 8, carry 6	72	{120}	D : [110, 111, 112, 113, 113, 114, 115, 116, 117, 117] λ : 129
117	Product by 8, carry 7	73	{136}	D : [110, 111, 112, 113, 113, 114, 115, 116, 117, 117] λ : 129
118	Erase x, bottom factor 9	36	{81}	x : 119
119	First product digit, factor 9	74	{82}	D : [120, 120, 121, 122, 123, 124, 125, 126, 127, 128]
120	Product by 9, carry 0	74	{82}	D : [120, 120, 121, 122, 123, 124, 125, 126, 127, 128] λ : 129
121	Product by 9, carry 1	75	{151}	D : [120, 121, 121, 122, 123, 124, 125, 126, 127, 128] λ : 129
122	Product by 9, carry 2	76	{177}	D : [120, 121, 122, 122, 123, 124, 125, 126, 127, 128] λ : 129
123	Product by 9, carry 3	77	{154}	D : [120, 121, 122, 123, 123, 124, 125, 126, 127, 128] λ : 129
124	Product by 9, carry 4	78	{176}	D : [120, 121, 122, 123, 124, 124, 125, 126, 127, 128] λ : 129
125	Product by 9, carry 5	79	{83}	D : [120, 121, 122, 123, 124, 125, 125, 126, 127, 128] λ : 129
126	Product by 9, carry 6	80	{156}	D : [120, 121, 122, 123, 124, 125, 126, 126, 127, 128] λ : 129
127	Product by 9, carry 7	81	{152}	D : [120, 121, 122, 123, 124, 125, 126, 127, 127, 128] λ : 129
128	Product by 9, carry 8	82	{98}	D : [120, 121, 122, 123, 124, 125, 126, 127, 128, 128] λ : 129
129	Scan completed product	83	{17}	D : 129 λ : 130
130	Inspect row above	84	{18}	D : 131 λ : 155
131	Cross upper operator	85	{19}	x : 132
132	First upper factor digit	85	{20}	D : 133
133	Scan upper factor prefix	85	{20}	D : 133 λ : 134
134	Select upper factor	26	{21}	D : [135, 137, 139, 141, 143, 145, 147, 149, 151, 153]
135	First prefix erasure, factor 0	27	{60}	D : 136
136	Later prefix erasure, factor 0	27	{61}	D : 136 x : 64
137	First prefix erasure, factor 1	28	{22}	D : 138
138	Later prefix erasure, factor 1	28	{23}	D : 138 x : 67
139	First prefix erasure, factor 2	29	{71}	D : 140
140	Later prefix erasure, factor 2	29	{72}	D : 140 x : 70
141	First prefix erasure, factor 3	30	{46}	D : 142
142	Later prefix erasure, factor 3	30	{47}	D : 142 x : 74
143	First prefix erasure, factor 4	31	{92}	D : 144
144	Later prefix erasure, factor 4	31	{93}	D : 144 x : 79
145	First prefix erasure, factor 5	32	{84}	D : 146
146	Later prefix erasure, factor 5	32	{85}	D : 146 x : 85
147	First prefix erasure, factor 6	33	{140}	D : 148
148	Later prefix erasure, factor 6	33	{141}	D : 148 x : 92
149	First prefix erasure, factor 7	34	{160}	D : 150
150	Later prefix erasure, factor 7	34	{161}	D : 150 x : 100
151	First prefix erasure, factor 8	35	{118}	D : 152
152	Later prefix erasure, factor 8	35	{119}	D : 152 x : 109
153	First prefix erasure, factor 9	36	{96}	D : 154

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Simulation Certificate (Continued)

i	Phase	g_i	Q_i	Read : next phase
154	Later prefix erasure, factor 9	36	{97}	D : 154 x : 119
155	Start summation	86	{25}	λ : 156
156	First initial left digit	87	{26}	D : 157
157	Continue initial left scan	87	{26}	D : 157 λ : 158
158	Inspect fresh lower row	88	{27}	0:8 : 159 λ : 205
159	First digit after leading cell	89	{28}	D : 160
160	Scan lower product	89	{28}	D : 160 λ : 161
161	First upper alignment digit	104	{29}	D : 162
162	Continue upper alignment	104	{30}	D : 162 λ : 163
163	Read upper units digit	90	{31}	D : [164, 165, 166, 167, 168, 169, 170, 171, 172, 173]
164	First lower update, value 0	92	{63}	λ : 174
165	First lower update, value 1	93	{99}	λ : 174
166	First lower update, value 2	94	{138}	λ : 174
167	First lower update, value 3	95	{32}	λ : 174
168	First lower update, value 4	96	{73}	λ : 174
169	First lower update, value 5	97	{133}	λ : 174
170	First lower update, value 6	98	{94}	λ : 174
171	First lower update, value 7	99	{162}	λ : 174
172	First lower update, value 8	100	{49}	λ : 174
173	First lower update, value 9	101	{146}	λ : 174
174	Return after lower blank	91	{33}	λ : 175
175	Read next upper digit	90	{34}	D : [178, 179, 180, 181, 182, 183, 184, 185, 186, 187]
176	Return with carry zero	91	{36}	λ : 177
177	Read upper, carry zero	90	{37}	D : [178, 179, 180, 181, 182, 183, 184, 185, 186, 187] λ : 200
178	Add 0 below	92	{35}	D : 176 λ : 174
179	Add 1 below	93	{50}	D : [176, 176, 176, 176, 176, 176, 176, 176, 176, 188] λ : 174
180	Add 2 below	94	{122}	D : [176, 176, 176, 176, 176, 176, 176, 176, 188, 188] λ : 174
181	Add 3 below	95	{86}	D : [176, 176, 176, 176, 176, 176, 176, 188, 188, 188] λ : 174
182	Add 4 below	96	{74}	D : [176, 176, 176, 176, 176, 176, 188, 188, 188, 188] λ : 174
183	Add 5 below	97	{139}	D : [176, 176, 176, 176, 176, 188, 188, 188, 188, 188] λ : 174
184	Add 6 below	98	{157}	D : [176, 176, 176, 176, 188, 188, 188, 188, 188, 188] λ : 174
185	Add 7 below	99	{175}	D : [176, 176, 176, 188, 188, 188, 188, 188, 188, 188] λ : 174
186	Add 8 below	100	{100}	D : [176, 176, 188, 188, 188, 188, 188, 188, 188, 188] λ : 174
187	Add 9 below	101	{126}	D : [176, 188, 188, 188, 188, 188, 188, 188, 188, 188] λ : 174
188	Return with carry one	103	{101}	λ : 189
189	Read upper, carry one	102	{102}	D : [190, 191, 192, 193, 194, 195, 196, 197, 198, 199] λ : 202
190	Add 1 inside lower block	93	{50}	D : [176, 176, 176, 176, 176, 176, 176, 176, 176, 188]
191	Add 2 inside lower block	94	{122}	D : [176, 176, 176, 176, 176, 176, 176, 176, 188, 188]
192	Add 3 inside lower block	95	{86}	D : [176, 176, 176, 176, 176, 176, 176, 188, 188, 188]
193	Add 4 inside lower block	96	{74}	D : [176, 176, 176, 176, 176, 176, 188, 188, 188, 188]
194	Add 5 inside lower block	97	{139}	D : [176, 176, 176, 176, 176, 188, 188, 188, 188, 188]
195	Add 6 inside lower block	98	{157}	D : [176, 176, 176, 176, 188, 188, 188, 188, 188, 188]
196	Add 7 inside lower block	99	{175}	D : [176, 176, 176, 188, 188, 188, 188, 188, 188, 188]
197	Add 8 inside lower block	100	{100}	D : [176, 176, 188, 188, 188, 188, 188, 188, 188, 188]
198	Add 9 inside lower block	101	{126}	D : [176, 188, 188, 188, 188, 188, 188, 188, 188, 188]

Continued on the next page

Simulation Certificate (Continued)

i	Phase	g_i	Q_i	Read : next phase
199	Add 10 inside lower block	105	{180}	D : 188
200	Preserve leading digit	87	{26}	0:8 : 201
201	Blank left of merged row	87	{26, 106}	λ : 158
202	Increment leading digit	93	{103}	0:8 : 203
203	Return after final carry	91	{104}	λ : 204
204	Step down left of sum	90	{105}	λ : 201
205	Emit final stop	106	{38}	λ : 206
206	Halted	107	{181}	–

5 Equivalence Proof

Proof of Theorem 1. Initially both executions have the same grid and head; phase 0 relates the two initial control states. Suppose the relation holds before a step. Lemma 5 supplies a permitted read symbol x. Equal grids and heads give the same read symbol to both controllers. Equation (4) gives the same write and motion and relates the successor states through phase j. The next grids and heads therefore agree.

Induction establishes equality of every finite execution prefix. The halt condition permits termination exactly on entry to phase 206. The expert terminates by the finite copy, product, and merge arguments; the extracted controller therefore terminates on the same step. The complete PTOs and final configurations agree. Lemma 3 supplies the final padded product and head position. □

Undefined Cells. The partial controller leaves $182 \cdot 12 - 1{,}271 = 913$ entries undefined. None occurs on a valid multiplication: protocol coverage and (4) supply a defined entry at every possible execution step. For example, controller state 0 scans the left operand and never reads a blank, whereas state 1 scans the right operand and handles the right blank. Both states correspond to expert state 0; merging these semantic roles during inspection would incorrectly demand a blank entry at controller state 0. The phase relation retains the necessary context.

6 Verification and Artifacts

Verification. Run `python3 verify_equivalence.py --quick` from this directory. The checker verifies fingerprints, CSV/JSON equality, equality of the expert export with `mult_grid(10)`, and every obligation in (4). The checked phase relation matches every row of the certificate. Omitting `--quick` additionally replays all 2,000 extraction inputs and the 100 saved pairs of 100-digit operands on separate autonomous grids. The 180,510 extraction actions and 15,114,148 long-input actions agree exactly; final grids, heads, and padded arithmetic products also agree. The invariants and induction establish the unbounded claim.

Artifact Identity. The title page gives the repository-relative case directory. This proof directory contains the controller and expert exports, reference code, protocol, checker, and saved verification report. The verification script stores or reads SHA-256 checksums for these artifacts. A checksum identifies the exact bytes of a file; the checker compares the stored and computed values to detect changes. These checks bind the certificate to the supplied exports. The invariants, protocol coverage, and local transition identity establish equivalence.